

SIMATS ENGINEERING
CO2_ASSESSMENT TOOL3_Application Based Questions

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

Name	Shaik Ashraf
Register Number	192324088

COURSE OUTCOME COVERED

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I Shaik Ashraf/192324088 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____



Assessment Tasks Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.
2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.
3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge

servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.
5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments.
Provide an example of a microservices-based application.

Application Based Questions Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity

Question 1 — Auto-Scaling & Load Balancing for Seasonal Traffic Spikes

Scenario

A company experiences unpredictable traffic spikes during seasonal sales and needs its infrastructure to handle demand efficiently without downtime.

How Auto-Scaling and Load Balancing Handle the Demand

- Auto-scaling automatically adds or removes compute instances (VMs/containers) based on real-time metrics such as CPU utilization, memory usage, or request count, so capacity always matches demand.
- Horizontal scaling (scaling out) is preferred over vertical scaling in the cloud because new instances can be provisioned within minutes to absorb sudden spikes.

How This Reduces Downtime & Improves User Experience

Benefit	Explanation
No single point of failure	Traffic is spread across multiple instances, so one server crashing does not take the whole application down.
Faster response times	Additional capacity prevents queueing/timeouts during flash sales, keeping page-load and checkout latency low.
High availability	Auto-healing replaces unhealthy instances automatically, maintaining near-continuous uptime.
Cost efficiency	Resources scale down after the spike, so the company pays only for what it uses (pay-as-you-go).

Real-World E-Commerce Example

During events like Amazon's Great Indian Festival or Flipkart's Big Billion Days, traffic can surge 10–20x within minutes.

Question 2 — Secure Cloud Backup Storage

Scenario

An organization wants to store large volumes of backup data securely in the cloud.

Durability & Data Redundancy

- Cloud storage services (e.g., Amazon S3, Azure Blob Storage, Google Cloud Storage) replicate every object across multiple physical devices and, often, multiple data centers/Availability Zones.
- Redundancy models include local redundancy (same data center), zone redundancy (across AZs), and geo-redundancy (across regions), letting the organization choose the level of protection needed.

Encryption & Access Control for Sensitive Data

- Encryption at rest (AES-256) protects stored backup files even if physical storage media is compromised.
- Encryption in transit (TLS/SSL) secures data while it is being uploaded or restored.

Cost Advantages vs Traditional Storage

Factor	Traditional On-Premise Storage	Cloud Storage
Capital expenditure	High upfront cost for hardware	Pay-as-you-go, no upfront hardware cost
Scalability	Manual procurement, slow	Elastic — scales instantly on demand
Maintenance	In-house staff & hardware upkeep	Managed by cloud provider
Redundancy	Costly to replicate across sites	Built-in multi-zone/region replication
Storage tiers	Fixed cost regardless of access frequency	Tiered pricing (hot/cool/archive) reduces cost for rarely accessed backups

Disaster Recovery Use Case

A bank stores daily encrypted backups of transaction databases in a cloud storage bucket with cross-region replication.

Question 3 — Global App Deployment with CDNs

Scenario

A mobile app company plans to deploy its application globally and needs low latency for all users.

How CDNs Improve Performance & Reduce Latency

- A Content Delivery Network (CDN) caches static and semi-static content (images, videos, scripts, API responses) on a distributed network of edge servers/Points of Presence (PoPs) located close to end users.
- Requests are served from the nearest edge location instead of a single origin server, drastically cutting round-trip time.

How Edge Servers Enhance User Experience Across Regions

Region Challenge	Edge Server Benefit
High network hops to origin	Content served locally, cutting latency from hundreds of ms to tens of ms
Regional traffic spikes	Load is distributed across many edge nodes instead of one origin
Unreliable long-distance links	Fewer network segments involved, reducing packet loss and jitter
Compliance/data residency	Some CDNs allow region-specific caching to meet local regulations

How Cloud Providers Support Global Availability

- Cloud providers offer multi-region deployment — running application instances in several geographic regions so users connect to the closest region.
- Global load balancers (e.g., AWS Global Accelerator, Google Cloud Load Balancing) route users to the nearest healthy region automatically.

Real-Time Streaming / Gaming Example

A live video streaming platform uses a CDN to cache video segments at edge locations worldwide, so a viewer in Chennai and one in Toronto both stream from a nearby edge node with minimal buffering.

Question 4 — Compliance & Security for Financial Institutions

Scenario

A financial institution requires strict regulatory compliance and data security in its cloud usage.

How Cloud Providers Support Regulatory Compliance & Auditing

- Major providers maintain compliance certifications (ISO 27001, SOC 2, PCI-DSS, GDPR, RBI/local financial regulations) and share responsibility under the shared responsibility model.
- Compliance dashboards (AWS Artifact, Azure Compliance Manager) give auditors ready access to certification reports and control mappings.

Role of Logging, Monitoring & Encryption

Control	Security Role
Logging (e.g., AWS CloudTrail)	Records every API call/user action for forensic investigation and audit trails
Monitoring (e.g., CloudWatch, SIEM)	Detects anomalies such as unusual login patterns or data exfiltration in real time
Encryption at rest & in transit	Protects transaction data and customer PII from unauthorized access
Key management (HSM/KMS)	Ensures only authorized services can decrypt sensitive financial records

Ensuring Data Sovereignty

- Selecting a cloud region within the required country/jurisdiction so data never leaves national borders unless permitted.
- Using data residency and geo-fencing controls to restrict where backups and replicas can be stored.
- Applying contractual data processing agreements (DPAs) with the cloud provider that specify legal jurisdiction and government-access limitations.

Example — Sensitive Financial Transactions

A bank processing card transactions stores transaction logs in a cloud region located within its home country to satisfy data sovereignty rules

Question 5 — Containerization for Application Deployment

Scenario

A development team wants to adopt containerization for application deployment.

Containers vs Virtual Machines

Aspect	Virtual Machines	Containers
Architecture	Each VM includes a full guest OS on top of a hypervisor	Containers share the host OS kernel, isolated via namespaces/cgroups
Startup time	Minutes (boots a full OS)	Seconds (just starts the process)
Resource usage	Heavier — full OS overhead per instance	Lightweight — only app + dependencies packaged
Density per host	Fewer VMs per physical server	Many more containers per server
Isolation strength	Strong (hardware-level via hypervisor)	Process-level isolation (slightly weaker than VMs)

How Orchestration (Kubernetes) Simplifies Deployment & Scaling

- Automated scheduling places containers on appropriate nodes based on resource availability.
- Self-healing: Kubernetes automatically restarts failed containers and replaces unhealthy pods.

Portability Benefits Across Cloud Environments

- A container image bundles the **application, its runtime, libraries, and configuration**, so it runs identically on any environment with a container runtime ("build once, run anywhere").

Microservices Example

An e-commerce platform splits its monolith into microservices — catalog, cart, payment, and recommendation services — each packaged as its own Docker container and deployed on a Kubernetes.